

Independent GenAI Model Validation Report

Prototype Retrieval-Augmented Generation (RAG) System

Portfolio Demonstration

Version 1.0

Table of Contents

- 1 Executive Summary
- 2 Document Control
- 3 Model Inventory
- 4 Business Context
- 5 Intended Use
- 6 System Architecture
- 7 Data Sources
- 8 Model Development
- 9 Validation Scope
- 10 Validation Methodology
- 11 Evaluation Metrics
- 12 Model Evaluation Results

1. Executive Summary

This report documents the independent validation of a Retrieval-Augmented Generation (RAG) prototype developed using open-source technologies. The objective was to evaluate retrieval quality, response grounding and generated answer quality using practical evaluation metrics. The document is intentionally structured to resemble a simplified banking model validation report.

2. Document Control

Document Owner: Raghav Mittal

Validation Type: Independent Prototype Review

Status: Draft

Business Area: GenAI / Model Risk Management

Purpose: Demonstrate understanding of RAG architecture and validation.

3. Model Inventory

Model Category: Retrieval-Augmented Generation

Knowledge Base: Banking and Basel regulatory PDFs

Embedding Model (Retrieval): BAAI/bge-small-en-v1.5

Embedding Model (Evaluation): all-MiniLM-L6-v2

Vector Store: FAISS

Frameworks: LangChain, Hugging Face

Execution Environment: Google Colab

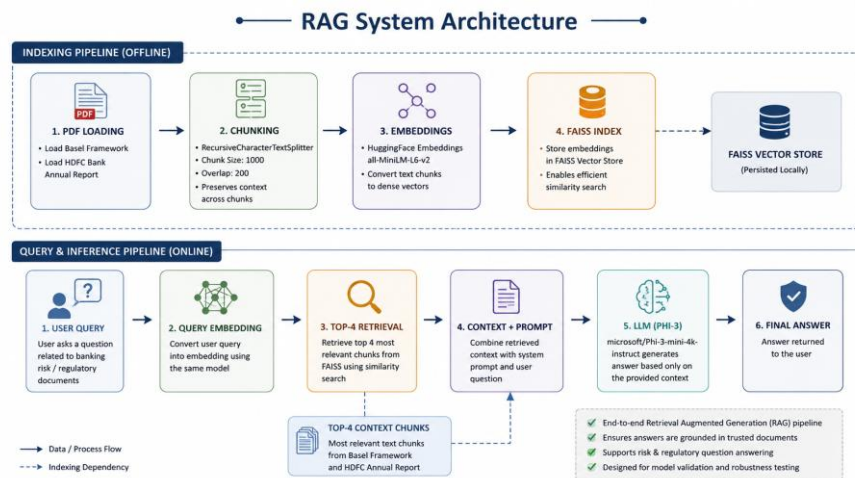
4. Business Context

Large Language Models can hallucinate because they generate statistically probable text rather than verifying facts. RAG reduces this risk by retrieving relevant enterprise documents before generation.

5. Intended Use

Internal knowledge assistant capable of answering questions from banking documents while grounding responses in approved source material.

6. System Architecture



7. Data Sources

Primary sources included Basel operational risk documentation and HDFC annual reports. PDFs were parsed using PyPDFLoader before preprocessing.

8. Model Development

Documents were split into 800-character chunks with 150-character overlap. Each chunk was embedded and indexed inside FAISS. User queries were embedded using the same embedding model and matched using cosine similarity. The four most relevant chunks were supplied to the language model.

9. Validation Scope

Included:

- Retrieval evaluation
- Response quality evaluation
- Hallucination assessment

Excluded:

- Bias
- Toxicity
- Prompt injection
- Fairness
- Production monitoring

10. Validation Methodology

Original intent was to use RAGAS. Due to API limitations, proxy metrics based on sentence embeddings and cosine similarity were implemented.

11. Evaluation Metrics

Metric	Validation Objective	Methodology	Interpretation
Retrieval Accuracy	Assess the effectiveness of the retrieval component in identifying document chunks relevant to the user's query.	Percentage of evaluation questions for which the retrieved context contained the information required to answer the question correctly.	Higher values indicate stronger retrieval performance and improved grounding of generated responses.
Context Relevance	Assess whether the retriever returned information relevant to the generated response.	Cosine similarity between the generated answer embedding and the retrieved context embedding.	Higher values indicate that the response is well grounded in the retrieved documents.

Answer Relevance	Measure whether the generated answer addresses the user's question.	Cosine similarity between the question embedding and the generated answer embedding.	Higher similarity indicates the response remains focused on the user's query.
Answer Correctness	Evaluate semantic correctness against an expected reference response.	Cosine similarity between the generated answer and the manually curated reference answer.	Higher similarity suggests greater semantic agreement with the expected response.
Hallucination Rate	Identify responses that are insufficiently supported by retrieved evidence.	Responses with Context Relevance < 0.70 are flagged as potential hallucinations and manually reviewed.	Lower hallucination rates indicate stronger grounding and improved reliability.

12 . Model Evaluation Results

The RAG system was evaluated using a benchmark dataset of twenty domain-specific banking risk questions. Performance was assessed across retrieval quality, semantic correctness, answer relevance, and grounding. The objective was to determine whether the retrieval pipeline consistently identified relevant document context and whether the Large Language Model generated responses supported by that context.

Evaluation Summary

Metric	Result	Assessment
Retrieval Accuracy	65.00%	Moderate
Answer Correctness	0.624	Moderate
Answer Relevance	0.554	Moderate
Context Relevance	0.724	Good
Hallucination Rate	35.00%	Requires Improvement

Retrieval Accuracy – 65.00%

The retriever successfully identified relevant document chunks for approximately two-thirds of the evaluation questions. While this demonstrates that the vector database is

capable of locating appropriate regulatory content, additional improvements in chunking strategy, embedding selection, or retrieval configuration may improve retrieval precision.

Answer Correctness – 0.624

Generated responses demonstrated moderate semantic alignment with the manually curated reference answers. The results indicate that the model was generally able to capture the key concepts from the retrieved documents, although responses occasionally lacked precision or omitted important details.

Answer Relevance – 0.554

This metric indicates that responses were only moderately aligned with the intent of the user's question. In several cases, the language model produced broader summaries of the retrieved context rather than directly answering the specific question posed. This behaviour is characteristic of smaller instruction-tuned language models operating with limited prompt constraints.

Context Relevance – 0.724

The generated responses exhibited good semantic alignment with the retrieved document context, suggesting that the RAG architecture generally succeeded in grounding model outputs in the supporting evidence rather than relying entirely on the model's internal knowledge. This is a positive indicator that the retrieval component is functioning as intended.

Hallucination Rate – 35.00%

Approximately one-third of the evaluated responses were identified as potential hallucinations based on the predefined context relevance threshold. These responses require manual review to determine whether they represent unsupported content, incomplete grounding, or retrieval failures. While the observed hallucination rate is higher than desirable for a production-grade deployment, it is not unexpected for a proof-of-concept implementation using a lightweight locally hosted language model without advanced guardrails or response verification mechanisms.

Overall Assessment

The validation results indicate that the Retrieval-Augmented Generation pipeline successfully grounds a majority of generated responses in the retrieved source documents. The relatively strong Context Relevance score demonstrates that the retrieval mechanism is capable of identifying semantically relevant evidence from the indexed document corpus.

However, the lower Answer Relevance and Answer Correctness scores suggest that the language model occasionally generates responses that are broader than required or insufficiently focused on the user's question. This behaviour contributes to the observed Hallucination Rate of 35%, highlighting the need for stronger prompt engineering,

enhanced retrieval strategies, or more capable instruction-tuned language models.

Overall, the proof-of-concept demonstrates that the RAG architecture is technically sound and suitable for domain-specific question answering, while also identifying clear opportunities for improvement before deployment in a production environment.

Validation Conclusion

Based on the evaluation performed, the model is assessed as acceptable for research and proof-of-concept purposes but not yet suitable for production deployment without additional controls. Recommended improvements include refinement of the retrieval pipeline, enhanced prompt engineering, implementation of confidence-based response filtering, and incorporation of additional robustness testing, including prompt injection and adversarial input evaluation.
